In [1]:
```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```python
# Set plotting style for better visuals
sns.set_theme(style="whitegrid")
plt.rcParams["figure.figsize"] = (10, 6)
```

In [5]:
```python
# 1. LOAD THE DATASET
df = pd.read_csv(r"C:\Users\Admin\Downloads\titanic.csv")
```

In [7]:
```python
# Display basic info
print(f"Dataset Shape: {df.shape}")
print("\nFirst 5 Rows:")
print(df.head())
```

```
Dataset Shape: (891, 12)

First 5 Rows:
   PassengerId  Survived  Pclass  \
0            1         0       3   
1            2         1       1   
2            3         1       3   
3            4         1       1   
4            5         0       3   

                                                Name     Sex   Age  SibSp  \
0                            Braund, Mr. Owen Harris    male  22.0      1   
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1   
2                             Heikkinen, Miss. Laina  female  26.0      0   
3       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1   
4                           Allen, Mr. William Henry    male  35.0      0   

   Parch            Ticket     Fare Cabin Embarked  
0      0         A/5 21171   7.2500   NaN        S  
1      0          PC 17599  71.2833   C85        C  
2      0  STON/O2. 3101282   7.9250   NaN        S  
3      0            113803  53.1000  C123        S  
4      0            373450   8.0500   NaN        S  
```

In [8]:
```python
# 2. DATA INTEGRITY & MISSING VALUES
```

In [12]:
```python
# Missing Values:-
missing_data = df.isnull().sum()
print(missing_data[missing_data > 0])
```

```
Age         177
Cabin       687
Embarked      2
dtype: int64
```

In [11]:
```python
# Basic descriptive statistics
```

```python
print(df.describe())
```

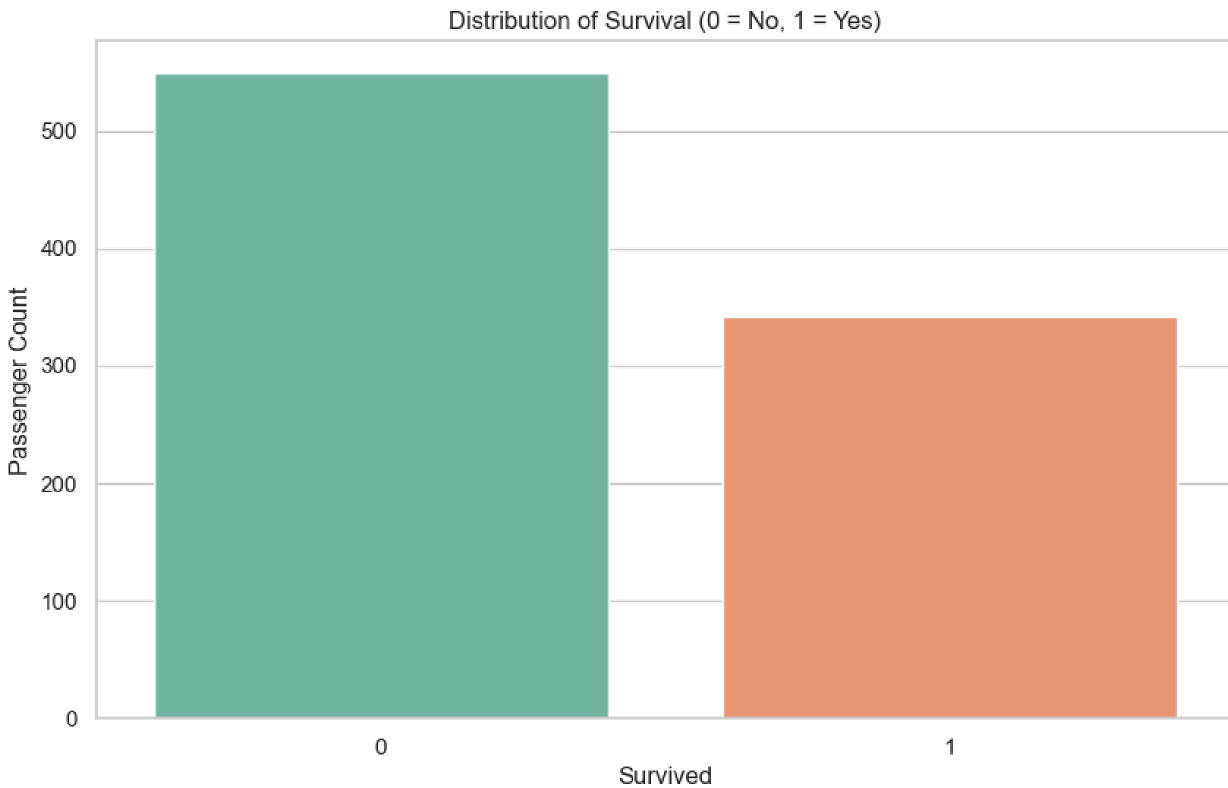
```
       PassengerId    Survived      Pclass         Age       SibSp  \
count   891.000000  891.000000  891.000000  714.000000  891.000000   
mean    446.000000    0.383838    2.308642   29.699118    0.523008   
std     257.353842    0.486592    0.836071   14.526497    1.102743   
min       1.000000    0.000000    1.000000    0.420000    0.000000   
25%     223.500000    0.000000    2.000000   20.125000    0.000000   
50%     446.000000    0.000000    3.000000   28.000000    0.000000   
75%     668.500000    1.000000    3.000000   38.000000    1.000000   
max     891.000000    1.000000    3.000000   80.000000    8.000000   

            Parch        Fare  
count  891.000000  891.000000  
mean     0.381594   32.204208  
std      0.806057   49.693429  
min      0.000000    0.000000  
25%      0.000000    7.910400  
50%      0.000000   14.454200  
75%      0.000000   31.000000  
max      6.000000  512.329200  
```

In [13]:

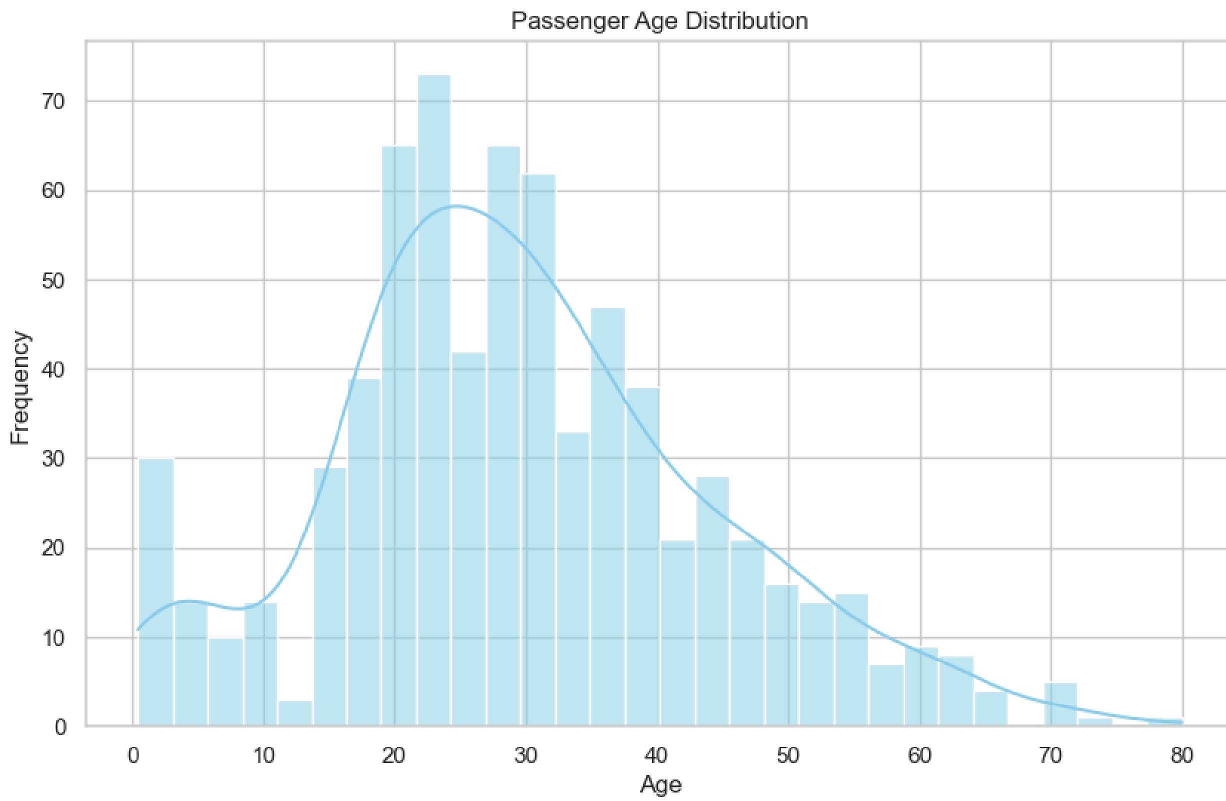
```python
# 3. DATA VISUALIZATION & ANALYSIS
```

In [19]:

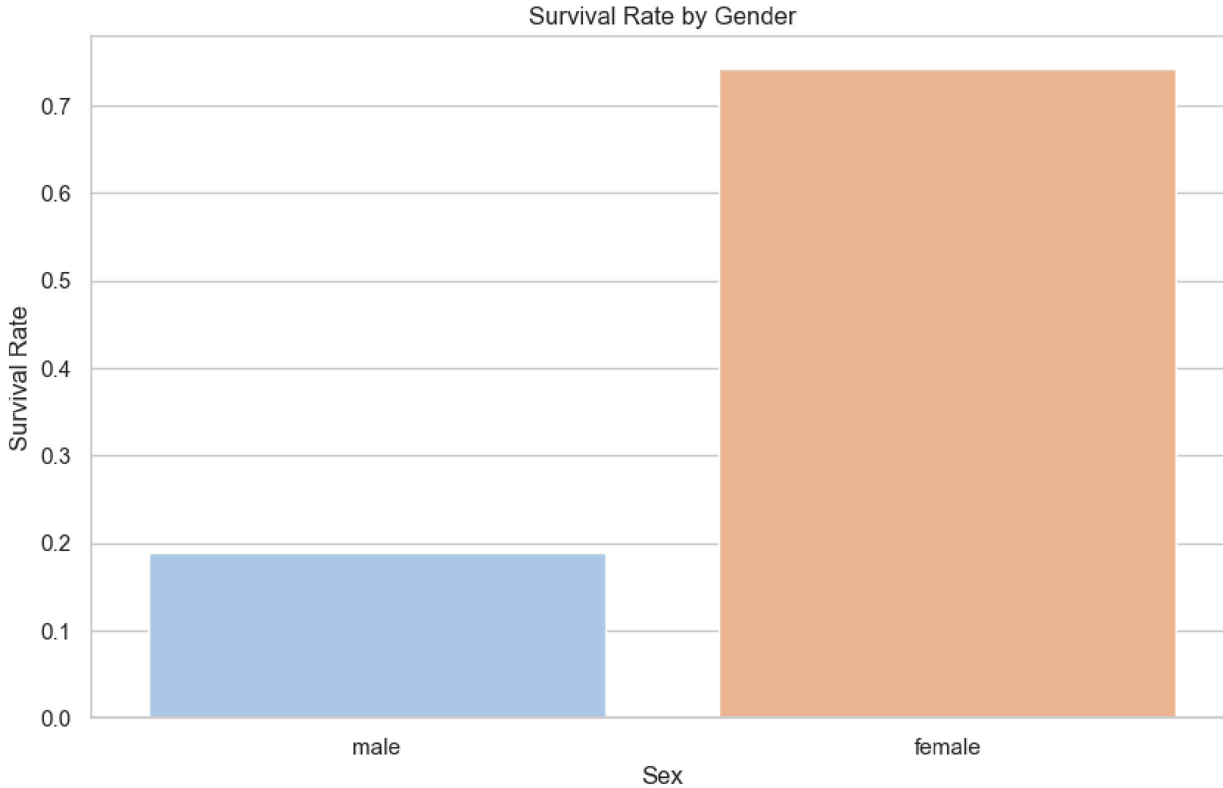
```python
# Plot 1: Survival Count (Bar Chart)
plt.figure()
sns.countplot(data=df, x="Survived", hue="Survived", palette="Set2", legend=Fa
plt.title("Distribution of Survival (0 = No, 1 = Yes)")
plt.xlabel("Survived")
plt.ylabel("Passenger Count")
plt.show()
```



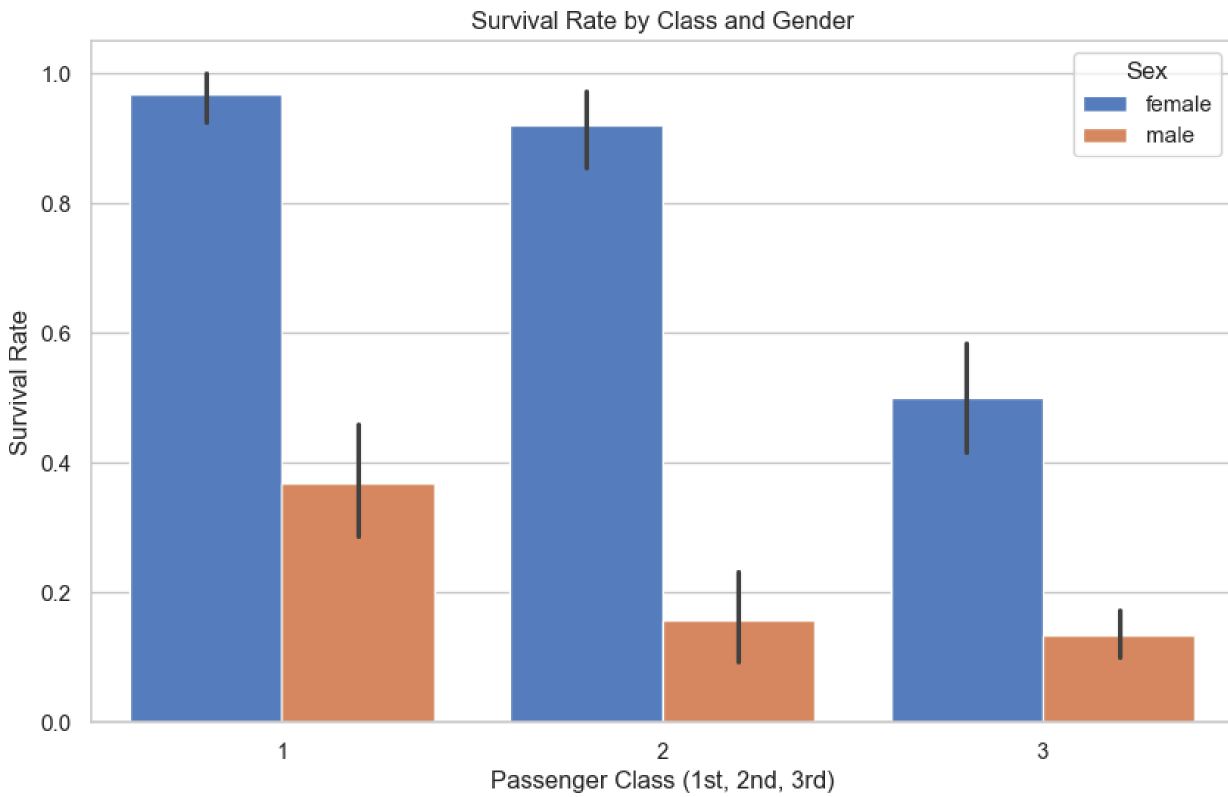
In [16]:
```
# Plot 2: Age Distribution (Histogram with KDE)
plt.figure()
sns.histplot(data=df, x="Age", kde=True, bins=30, color="skyblue")
plt.title("Passenger Age Distribution")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.show()
```




```
In [22]: # Plot 3: Survival Rate by Gender (Bar Chart)
plt.figure()
sns.barplot(data=df, x="Sex",hue="Sex", y="Survived", errorbar=None, palette="
plt.title("Survival Rate by Gender")
plt.ylabel("Survival Rate")
plt.show()
```



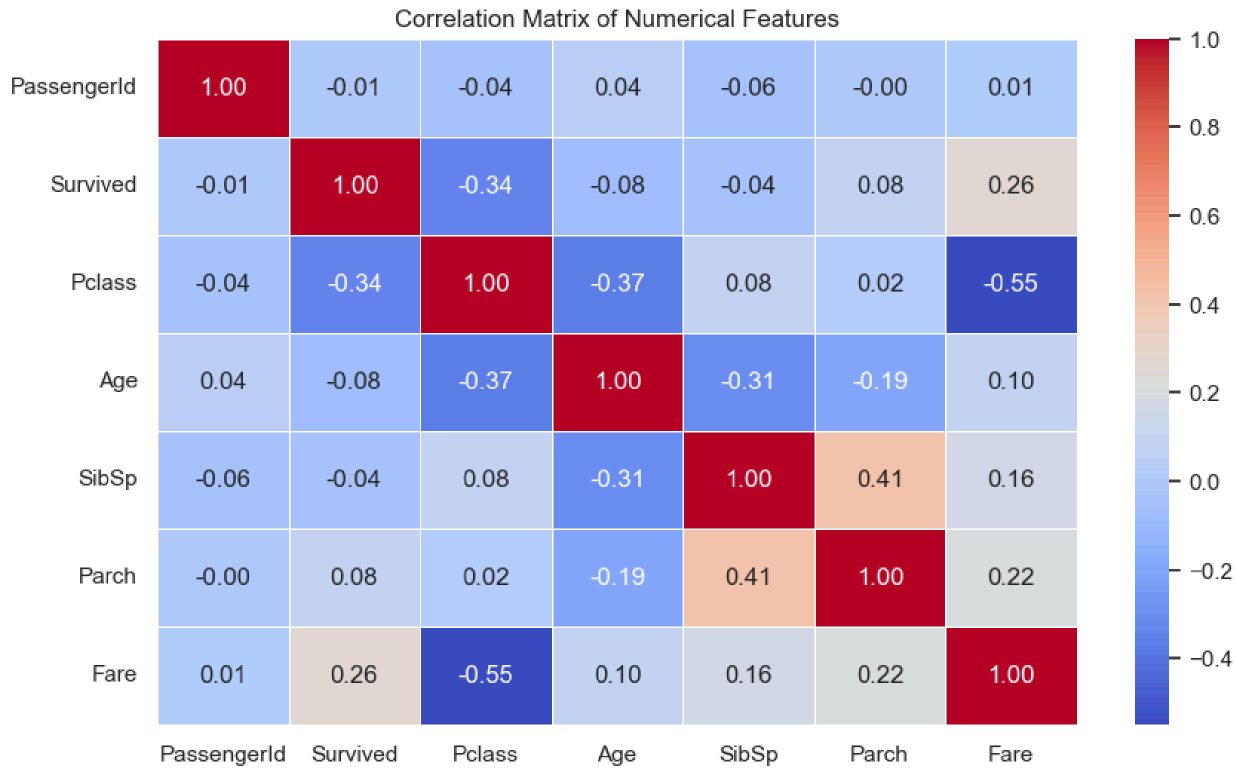
In [26]:
```
# Plot 4: Survival Rate by Passenger Class
plt.figure()
sns.barplot(data=df, x="Pclass", y="Survived", hue="Sex", palette="muted")
plt.title("Survival Rate by Class and Gender")
plt.ylabel("Survival Rate")
plt.xlabel("Passenger Class (1st, 2nd, 3rd)")
plt.show()
```



In [25]:
```
# Plot 5: Correlation Heatmap
plt.figure(figsize=(8, 6))

# Selecting only numerical columns for correlation
numerical_cols = df.select_dtypes(include=[np.number]).columns
correlation_matrix = df[numerical_cols].corr()
```

```
<Figure size 800x600 with 0 Axes>
```

In [27]:
```
sns.heatmap(correlation_matrix, annot=True, cmap="coolwarm", fmt=".2f", linewi
plt.title("Correlation Matrix of Numerical Features")
plt.show()
```



Correlation Matrix of Numerical Features
PassengerId
Survived
Pclass
Age
SibSp
Parch
Fare
1.00 -0.01 -0.04 0.04 -0.06 -0.00 0.01
-0.01 1.00 -0.34 -0.08 -0.04 0.08 0.26
-0.04 -0.34 1.00 -0.37 0.08 0.02 -0.55
0.04 -0.08 -0.37 1.00 -0.31 -0.19 0.10
-0.06 -0.04 0.08 -0.31 1.00 0.41 0.16
-0.00 0.08 0.02 -0.19 0.41 1.00 0.22
0.01 0.26 -0.55 0.10 0.16 0.22 1.00
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4


```
In [ ]:
```